

Reference excerpt 01-036

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of the relationship expressed in Eq. 4, can be tested by comparing the instantaneous velocity ($w(t)$) of the blood in the IJV calculated according to the equation shown above, with the instantaneous blood velocity ($w_D(t)$) measured with a Doppler scanner. However, the data and the results presented here are for illustration only and are not to be considered the result of a scientific study.

Pressure waves velocity calculation The delay ta_{QP} is measured over the JVP+ECG trace as shown in Fig. 2. The distance IG is measured on the volunteers chest. The velocity c is the ratio between IG and ta_{QP} .

Compliance calculation From the Moens-Korteweg equation we obtain the compliance per unit length: $C = CSA \times c^2$ (5) where CSA_x is the CSA of the IJV measured in G at the x wave.

Pressure gradient calculation The pressure inside the IJV is calculated from the instantaneous CSA as follows: $p(t) = \frac{1}{C} CSA(t)$ (6) substituting the expression of $p(t)$ obtained above into Eq. (2) we obtain the expression for the pressure gradient: $p_z = \frac{1}{C} \frac{dCSA}{dt}$ (7)

Flow velocity calculation $w(t, z)$ The function $w(t, z)$ is calculated by inserting into the Eq.(1) the expression for the pressure gradient found in Eq. (7). The mathematical details are given in [Sisini et al.(2015b)].

3 Figure 3: Calculated velocity trace together with the experimental one obtained by the Doppler examination.

RESULTS EXAMPLE Pressure waves velocity calculation The measured delay (ta_{QP}) was 0.14 s and